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Roll No

EE-6005(2)-CBGS

B.E. VI Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Digital Control System

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Discuss about sampling process and its frequency domain analysis.
b) Discuss sampling theorem for band pass signal.
2. a) Define Z transform. Show the mapping between S plane and Z plane.
b) Obtain the Z transform of $X(s) = \frac{1 - e^{-sT}}{s} \frac{1}{(s + a)^2}$.
3. Explain the following.
 - i) Digital PID controllers
 - ii) Velocity and position forms of digital PID controller
4. Discuss about the stability using Routh's test and Jury's test.
5. Discuss the root locus analysis for stability of discrete control system.

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6. Discuss the state variable representation in the Z domain.
7.
 - a) Explain the concept of controllability and observability of discrete time control system.
 - b) What is meant by performance index of an optimal control system?
8. Explain the following. (any two)
 - i) Sampled data control system
 - ii) Jordan transformation
 - iii) System stability
 - iv) State transition equation
